PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T40 Classified

Cross-session • chapter-organised candidate

4 classified items • 18 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Unit 1 • Chapter OL-T40 • 4 items • 18 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Angles in Polygons & Parallel Lines

Unit 1 • 18 marks • 4 item(s)

June 2023 | 4MA1-2H Q3 | 4 marks

Exploit parallel and equal sides in a special quadrilateral

January 2020 | 4MA1-2H Q4 | 4 marks

Solve an algebraic triangle-angle chain with reasons

June 2021 | 4MA1-2H Q5 | 5 marks

Complete a multi-step line-angle reason chain

May-Nov 2020 | 2H Q3 | 5 marks

Give standard reasons in a quadrilateral angle chain

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 14
4MA1-2H Q3

ORIGINAL QUESTION
4 marks

3 *ABCD* is a trapezium.

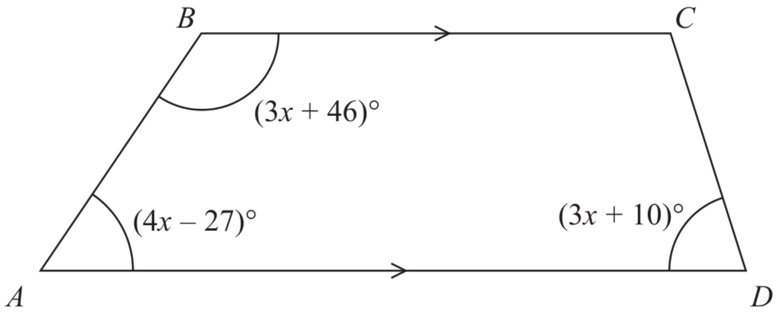

Diagram **NOT** accurately drawn

BC is parallel to *AD*

Find the size of the largest angle inside the trapezium.

.... °

Worked Solution 14

Largest angle in a trapezium

MODULAR UNIT 1
4 marks

Use parallel-line angles

$$(4x - 27) + (3x + 46) = 180 \Rightarrow 7x + 19 = 180 \Rightarrow x = 23.$$

Evaluate the angles

$$3x + 46 = 115, \quad 3x + 10 = 79, \quad 4x - 27 = 65; \text{ the remaining angle is } 101.$$

Final answer

$$115^\circ$$

4 The diagram shows a triangle.

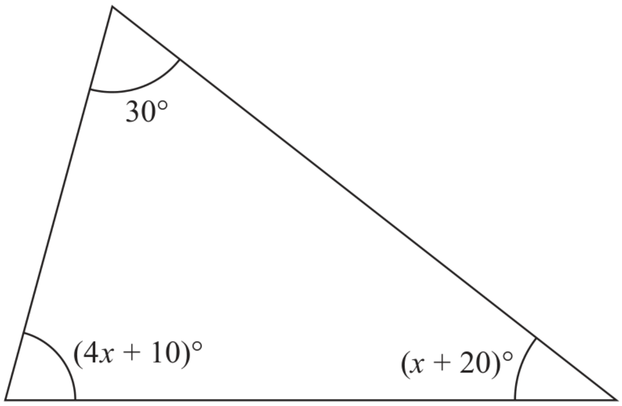

Diagram **NOT** accurately drawn

Work out the value of x.

$x =$ ....

Worked solution

January 2020 | Question 4 | 4 marks

Family: Use triangle angle properties • Variant: Solve an algebraic triangle-angle chain with reasons

Method / working

1. Triangle angles sum to 180°.
2. $30+(4x+10)+(x+20)=180$.
3. $5x+60=180$, so $5x=120$.
4. $x=24$.

Final answer: $x=24$

5

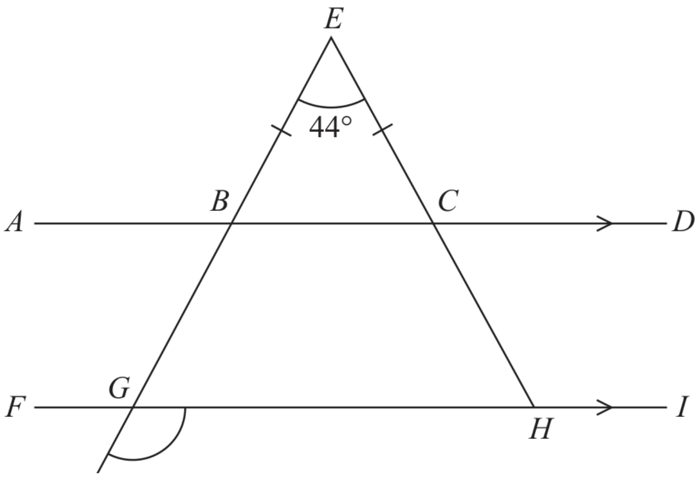

Diagram **NOT** accurately drawn

ABCD and *FGHI* are parallel straight lines.
EBGJ and *ECH* are straight lines.

$BE = CE$
Angle $BEC = 44°$

Work out the size of angle *JGH*.
Give a reason for each stage of your working.

.... °

Worked solution

June 2021 | Question 5 | 5 marks

Family: Use basic angle and line facts • Variant: Complete a multi-step line-angle reason chain

Method / working

1. BE=CE, so base angles of triangle BEC are (180-44)/2=68°.
2. Use the straight line at B: the adjacent angle is 180-68=112°.
3. Transfer the angle through the parallel lines using corresponding/alternate angles.
4. Use the straight-line relation at G to reach angleJGH=112°.
5. State the isosceles, straight-line, parallel and angle-sum reasons.

Final answer: 112°

May-Nov 2020 | 2H | Question 3 | 5 marks

Chapter OL-T40: Angles in Polygons & Parallel Lines

3

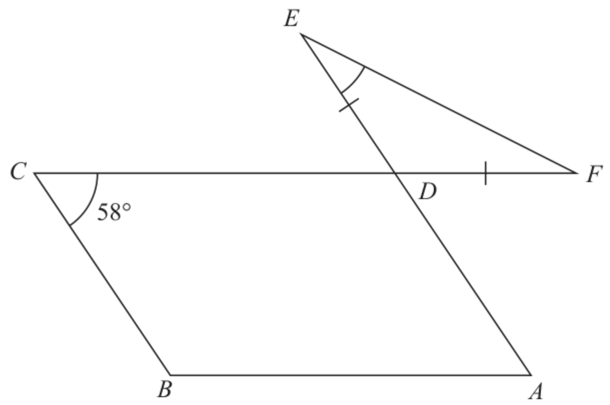

Diagram **NOT** accurately drawn

The diagram shows a parallelogram *ABCD* and an isosceles triangle *DEF* in which *DE* = *DF*

CDF and *ADE* are straight lines.
Angle *BCD* = 58°

Work out the size of angle *DEF*.
Give a reason for each stage of your working.

.... °

(Total for Question 3 is 5 marks)

Answer reference | May-Nov 2020 | Question 3

Family: Use quadrilateral properties • Variant: Give standard reasons in a quadrilateral angle chain

3	$ADC = 180 - 58$ $(= 122)$ **or** $EDF = 122$ **or** $CDE = 58$ **or** $ADF = 58$			M1	may be seen marked on the diagram
	e.g. $DEF = 58 \div 2$ **or** $DEF = (180 - 122) \div 2$			M1	complete method to find angle *DEF*
		29		A1	
			5	B2	dep on M2 for fully correct reasons for their method (B1 dep on M1 for one correct reason stated and used) e.g. Allied angles, co-interior angles, Alternate angles, Corresponding angles, Vertically opposite angles are equal (or Vertically opposite angles are equal), Angles on a straight line add up to 180°(or angles on a straight line add to 180°), Sum of two angles in a triangle are equal to opposite exterior angle, Angles in a triangle add up to 180°(or Angles in a triangle add up to 180°), Base angles in an isosceles triangle Angles in a quadrilateral add up to 360. (accept "4-sided shape"or parallelogram) Opposite angles of a parallelogram are equal
					Total 5 marks

Worked solution

May-Nov 2020 | Question 3 | 5 marks

Family: Use quadrilateral properties • Variant: Give standard reasons in a quadrilateral angle chain

Official final mark-scheme pages are embedded immediately after this question.